

LEVENT ERKÖK

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leventerkok

leventerkok.github.io

SUMMARY

Formal verification engineer with 20+ years of experience applying formal methods in the hardware industry. Spanning CPUs, GPUs, SoCs, and accelerators; covering floating-point circuits, multipliers, UMLA designs, crypto hardware amongst many other domains. Expertise in SAT/SMT solving, symbolic execution, model checking, and interactive theorem proving. Have led verification programs and built scalable methodologies to support them. Author and maintainer of SBV, an open-source Haskell framework for symbolic execution and SMT-based verification.

TECHNICAL EXPERTISE

Formal Verification Model Checking Symbolic Execution SAT/SMT Solving Theorem Proving CPU SoC

GPU Hardware Accelerators RTL Verification Property Checking Equivalence Checking Floating-Point HW

SystemVerilog Verilog Tcl Haskell Functional Programming Jasper FPV/C2RTL Hector

PROFESSIONAL EXPERIENCE

Principal Engineer

NVIDIA

August 2026 – Present

- Formal verification of Language Processing Units (LPUs).

Formal Verification Engineer

Apple

March 2019 – August 2026

- Led CPU and SoC formal verification efforts, defining verification strategy and execution.
- Proved correctness of SystemVerilog/Verilog RTL, including multiplier circuits and unified-multiplier designs, using Jasper FPV/C2RTL and Hector.
- Designed and implemented a security-focused DSL and its associated compiler chain for hyper-properties, compiling executable specifications to SystemVerilog with Jasper as the backend prover.

Formal Verification Engineer

Intel Corporation

August 2011 – March 2019

- Proved correctness of critical functionality in many-core microprocessors spanning client, server, and HPC product lines, using Jasper and Forte based tools.
- Verified floating-point circuitry, error-correction logic, and internal communication protocols.
- Co-authored IEEE Transactions on Computers work on high-radix division and square-root algorithms, with machine-checked correctness proofs, plus several other internal conference papers.

Member of Technical Staff

Galois, Inc.

December 2006 – July 2011

- Formal verification technology developer, combining functional programming, SAT/SMT solving, and interactive theorem proving.
- Core contributor to Cryptol, a DSL for specifying and verifying cryptographic algorithms across the high-assurance software stack; built an SMT-based proving engine for the language.
- Integrated automated solvers (MiniSat, Yices) into interactive theorem-proving workflows.

Research Scientist

Intel Corporation

📅 April 2006 – December 2006

📍 Hillsboro, Oregon

- Applied formal methods to trusted computing architectures in Intel's Trusted Platforms Lab.
- Developed functional-programming models of hypervisor behavior.
- Verified system-level design artifacts with SMV and SPIN model checking.

Senior Automation / Software Engineer

Intel Corporation

📅 October 2002 – April 2006

📍 Hillsboro, Oregon

- Designed automation infrastructure (primarily C++) for semiconductor manufacturing facilities.
- Used SMV model checking to specify and verify properties of controller scripts deployed in production fabs.

OPEN-SOURCE SOFTWARE

🌀 SBV— SMT-Based Verification

Author and Maintainer

- An open-source Haskell framework for symbolic execution and SMT-based verification.
- Provides an embedded DSL for expressing symbolic computations and correctness properties directly in Haskell, supporting both push-button proving and a lightweight theorem-proving interface.
- Interfaces Haskell to SMT solvers, automating reasoning about software and hardware properties.
- Implements symbolic execution, constraint solving, theorem-proving workflows, and code generation.

Additional Projects: SBV-Plugin (a GHC plugin for SMT-based verification of Haskell programs), HArduino (a Haskell interface for Arduino), and CrackNum (IEEE-754 encode/decode).

EDUCATION

Ph.D., Computer Science

OGL School of Science and Engineering, OHSU

2002

Value Recursion in Monadic Computations · Advisor: John Launchbury

M.S., Computer Science

The University of Texas at Austin

1998

M.S., Computer Engineering

Middle East Technical University, Ankara, Türkiye

1996

B.S., Computer Engineering

Middle East Technical University, Ankara, Türkiye

1994

SELECTED PUBLICATIONS

- W. E. Ferguson, J. Bingham, L. Erkök, J. R. Harrison, and J. Leslie-Hurd. Digit Serial Methods with Applications to Division and Square Root. *IEEE Transactions on Computers*, 2018.
- L. Erkök and N. Amjadi. Counting ECC Failures Using Formal Methods to Substantiate RAS Claims. *Intel Design and Test Technology Conference*, 2017.
- L. Erkök, M. Carlsson, and J. Matthews. Hardware/Software Co-Verification of Cryptographic Algorithms Using Cryptol. *FMCAD*, 2009.
- L. Bulwahn, A. Krauss, F. Haftmann, L. Erkök, and J. Matthews. Imperative Functional Programming with Isabelle/HOL. *TPHOLS*, 2008.
- L. Erkök and J. Matthews. Using Yices as an Automated Solver in Isabelle/HOL. *Automated Formal Methods*, 2008.
- L. Erkök and J. Launchbury. A Recursive Do for Haskell. *Haskell Workshop*, 2002.
- L. Erkök, J. Launchbury, and A. Moran. Semantics of Value Recursion for Monadic Input/Output. *Theoretical Informatics and Applications*, 2002.
- L. Erkök and J. Launchbury. Recursive Monadic Bindings. *ICFP*, 2000.

Complete publication list: leventerkok.github.io